

INTEGRAIS IMPORTANTES

C = CONSTATANTE

$$(i) \int du \rightarrow \int 1 \cdot du = \boxed{u + c}$$

$$(ii) \int u^n du \rightarrow \frac{d}{du}(u^n) = n \cdot u^{n-1} \therefore \int u^n du = \boxed{\frac{u^{n+1}}{n+1} + c}$$

$$(iii) \int a^u \cdot du \rightarrow \frac{d}{du}(a^u) = a^u \cdot \ln a \therefore \int a^u du = \boxed{\frac{a^u}{\ln a} + c}$$

$$(iv) \int \frac{du}{u} = \ln|u| + c$$

MÓDULO RESTRINGE AOS POSITIVOS

$$(v) \int e^u du = e^u + c$$

TRIGONOMÉTRICAS

$$(vi) \int \operatorname{sen} u du = -\operatorname{cos} u + c$$

$$(xi) \int \operatorname{csc} u \cdot \operatorname{cotg} u du = -\operatorname{csc} u + c$$

$$(vii) \int \operatorname{cos} u du = \operatorname{sen} u + c$$

$$(xii) \int \frac{du}{1+u^2} = \operatorname{arctg} u + c$$

$$(viii) \int \sec^2 u du = \operatorname{tg} u + c$$

$$(xiii) \int \frac{du}{\sqrt{1-u^2}} = \operatorname{arcsen} u + c$$

$$(ix) \int \operatorname{cosec}^2 u du = -\operatorname{cotg} u + c$$

$$(xiv) \int \frac{du}{u\sqrt{u^2-1}} = \operatorname{arcsec} u + c$$

$$(x) \int \sec u \cdot \operatorname{tg} u du = \sec u + c$$

$$(xv) \int \operatorname{tg} x dx = \ln|\sec x| + c$$

EXTRAS

$$\textcircled{i} \int \frac{du}{u^2+a^2} = \frac{1}{a} \operatorname{tg}^{-1}\left(\frac{x}{a}\right) + C$$